

Lab 1:

The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'Query' tab. The query ranks films by length in descending order and selects the top 3. The 'Data Output' tab shows the results of the query.

```
Query
```

```
4      c.name AS category_name,  
5      f.length,  
6      DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC) as rank  
7  FROM film f  
8  JOIN film_category fc ON f.film_id = fc.film_id  
9  JOIN category c ON fc.category_id = c.category_id  
10 )  
11 SELECT  
12     title,  
13     category_name,  
14     length,  
15     rank  
16 FROM RankedFilms  
17 WHERE rank <= 3;
```

Data Output

	title	category_name	length	rank
1	Worst Banger	Action	185	1
2	Dam Forrester	Action	185	1
3	Casualties Encino	Action	179	2
4	Quest Mussolini	Action	177	3
5	Pond Seattle	Animation	185	1
6	Gangs Pride	Animation	185	1
7	Theory Mermaid	Animation	184	2
8	Sons Interview	Animation	184	2
9	Lawless Vision	Animation	181	3
10	Wrong Behavior	Children	178	1
11	Fury Murder	Children	178	1

Total rows: 64 Query complete 00:00:00.147

Lab 2:

The screenshot shows the pgAdmin 4 interface with a SQL query executed in the 'Query' tab. The query calculates the growth percentage of revenue over time. The 'Data Output' tab shows the results of the query.

```
Query
```

```
27 ),  
28 RevenueWithLag AS (  
29     SELECT  
30         payment_month,  
31         current_revenue,  
32         LAG(current_revenue, 1) OVER (ORDER BY payment_month) AS previous_revenue  
33     FROM MonthlyRevenue  
34 )  
35 SELECT  
36     payment_month,  
37     current_revenue,  
38     previous_revenue,  
39     (current_revenue - previous_revenue) / previous_revenue * 100 AS growth_percentage  
40 FROM RevenueWithLag;
```

Data Output

	payment_month	current_revenue	previous_revenue	growth_percentage
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Total rows: 4 Query complete 00:00:00.074

Successfully run. Total query runtime: 74 msec. 4 rows affected.

Lab 3:

pgAdmin 4

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgres@PostgreSQL 18*

Query

```

((current_revenue - previous_revenue) / previous_revenue) * 100 AS growth_percentage
FROM RevenueWithLag;

SELECT
    customer_id,
    payment_id,
    payment_date,
    amount,
    SUM(amount) OVER (
        PARTITION BY customer_id
        ORDER BY payment_date
    ) AS clv_running_total
FROM payment
ORDER BY customer_id, payment_date;

```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 15

	customer_id	payment_id	payment_date	amount	clv_running_total
1	smallint	[PK] integer	timestamp without time zone	numeric (5,2)	numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92
9	1	22681	2007-03-02 14:05:18.996577	3.99	40.91
10	1	22682	2007-03-02 16:30:04.996577	0.99	41.90
11	1	22683	2007-03-17 11:06:20.996577	4.99	46.89

Total rows: 14596 Query complete 00:00:00.210 CRLF Ln 52, Col 36

Lab 4:

pgAdmin 4

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgres@PostgreSQL 18*

Query

```

last_name,
country,
total_spend,
RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) AS spend_rank
FROM CustomerSpend
)
SELECT
    customer_id,
    first_name,
    last_name,
    country,
    total_spend
FROM RankedCustomers
WHERE spend_rank = 1;

```

Data Output Messages Notifications

Showing rows: 1 to 108 Page No: 1 of 1

	customer_id	first_name	last_name	country	total_spend
1	integer	character varying (45)	character varying (45)	character varying (50)	numeric
1	218	Vera	Mccoy	Afghanistan	67.82
2	176	June	Carroll	Algeria	151.68
3	320	Anthony	Schwab	American Samoa	47.85
4	528	Claude	Herzog	Angola	93.80
5	381	Bobby	Boudreau	Anguilla	99.68
6	560	Jordan	Archuleta	Argentina	129.71
7	41	Stephanie	Mitchell	Armenia	118.75
8	266	Nora	Herrera	Austria	113.74
9	334	Raymond	Mcwhorter	Azerbaijan	108.75
10	590	Seth	Hannon	Bahrain	108.75
11	21	Michelle	Clark	Bangladesh	108.75

Total rows: 108 Query complete 00:00:00.122 CRLF Ln 86, Col 22

Successfully run. Total query runtime: 122 msec. 108 rows affected.